

Charu Datta Perur

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LinkedIn | Portfolio | Vellore, Tamil Nadu, India

Summary

Computer Science undergraduate (AI & Robotics specialization) with hands-on experience across **research** (prompt-routing systems, generative model evaluation) and **full-stack development**, currently deepening core **data structures, algorithms, and backend engineering** skills. Comfortable working across the stack, from distributed embedded systems to production-ready web interfaces.

Technical Skills

Languages: Java, C, C++, Python, JavaScript

Web: React.js, Next.js, Node.js, HTML5, CSS3

Tools: Git, Arduino IDE, UiPath

Other: React Native, GSAP, Embedded Systems, Team Leadership

Professional Experience

Centre of Advanced Data Science (CADS), VIT Chennai — Research Intern *May 2025 – Oct 2025*

- Co-authored **CRISP**, a prompt-routing system that scores the creativity level of incoming requests and forwards each to the most cost-effective generative model, achieving up to **35% savings** in inference spend.
- Assembled a 500-sample training set with **GPT/Gemini API** ratings and human annotation; evaluated **GAN, Transformer, and Diffusion** architectures head-to-head using CLIP Score.

Integra Global Solutions — Application Development Intern *May 2025 – July 2025*

- Owned UI development for a cross-platform mobile app — translated design specs into responsive, production-ready screens and shipped features that measurably improved user flow and load performance.

Projects

Smart RFID Access Control with Face Capture *Arduino Uno, 8051, ESP32-CAM, MFRC522*

- Distributed authentication logic across three microcontrollers — Arduino Uno handles RFID card classification and relays results over UART to an 8051, which drives the servo lock, I2C LCD, and LED/buzzer outputs.
- Solved stale-frame capture by triggering a full ESP32-CAM deinit/reinit cycle on each intrusion event; exposed the latest intruder photo over a self-hosted Wi-Fi endpoint with IPFS-ready architecture for future audit logging.

AutoNova Parts — Inventory Stock Alert System *UiPath, HTML/CSS/JS, jsPDF*

- Eliminated manual restocking by building a UiPath bot that polls the admin dashboard every 15 seconds, fires supplier emails via Gmail SMTP, sends WhatsApp confirmations, and clicks the ML restock button autonomously — zero human intervention after checkout.
- Engineered a browser-only ensemble forecasting engine (EMA + linear trend + velocity scoring) to predict 7-day part demand and auto-generate PDF purchase orders; synced admin and storefront views in real time using BroadcastChannel with no backend server.

Education

Vellore Institute of Technology, Chennai *August 2023 – August 2027*

Bachelor of Technology — Computer Science Engineering (AI & Robotics Specialization) **CGPA: 8.82/10**

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Databases, Operating Systems, Computer Networks, Robot Programming, Sensors & Actuators, Theory of Computation

Narayana Junior College *March 2023*

Intermediate (Class XII) — MPC, TSBIE **Marks: 970/1000**

Clubs & Activities

Linux Club, VIT Chennai — Event Marketing Member

- Drove outreach and promotions for club events, coordinating publicity campaigns and growing student engagement across campus.